

-
-
-
-
-

Chapter 8

Style Production Definition

Management Overview



Introduction

Production Definition Management, shown in the figure below, manages the Style Production database. This database contains:

- Extended production details of all the styles and materials used within Style Production.
- Information about resources (for example, labour skills and machines) that you use to combine materials into finished products.
- Facilities to cost products.

Style Inventory Management maintains information about styles and materials, both purchased and produced. Therefore, you must define in Style Inventory Management every style and material that you wish to use within Style Production.



Note: Before implementing Style Production within your system, make sure you have a clear understanding of the database, its links to Style Inventory Management and the company profile.

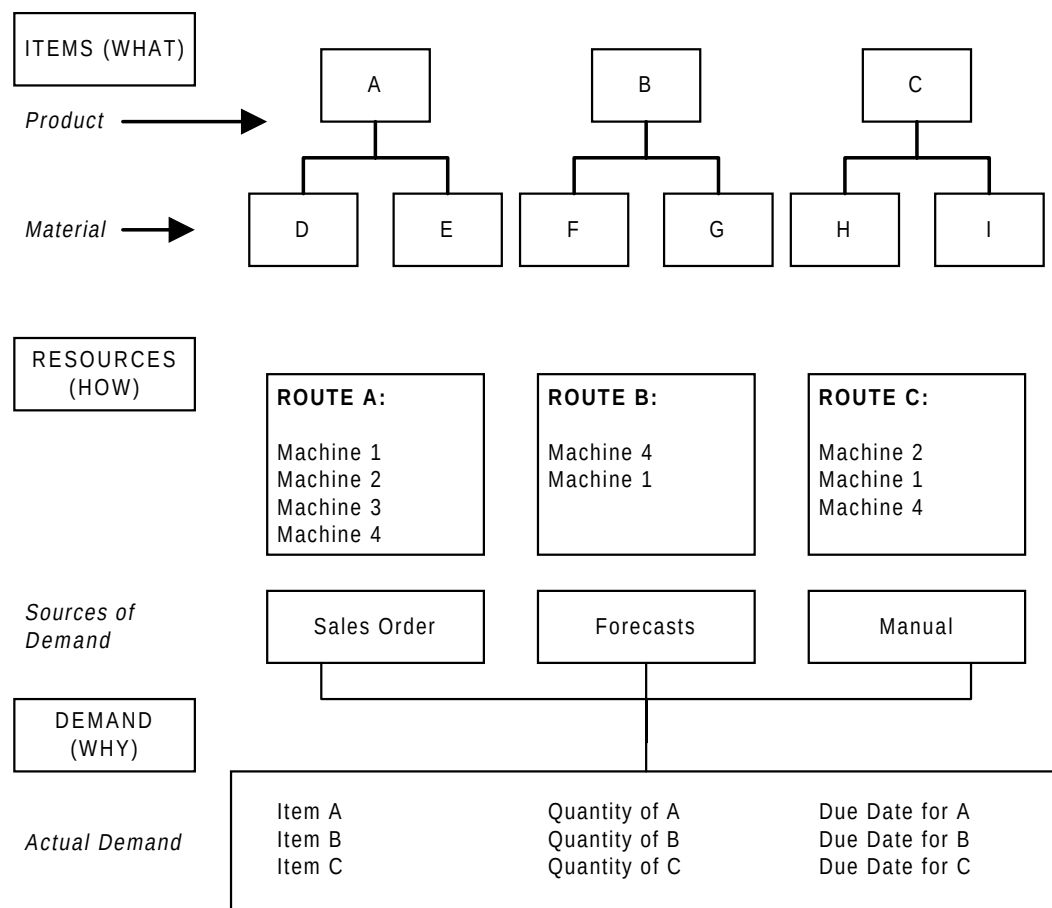


Figure: Production Definition Management

Resources

Resources are classified as:

- Machines, which you can group into work centres. The assigned cost centre defines the cost basis of each machine.
- Departments.
- Labour skills and profiles.
- Operators and teams.
- Subcontractors.

Company Profile

Style Production, in common with all JBA Business applications, operates within a multi-company environment.

For each production company, you need to create a company profile to:

- Set the basic system defaults and operation policies relevant to the company.
- Define the costing elements.
- Define base calculation parameters used within Style Production.



Note: Access to the company profile is usually restricted to system managers and implementation project managers who require a comprehensive understanding of the implications of each parameter setting. However, all users need to be aware of the purpose and features of the Company Profile.

In addition, to specify how you wish to carry out production at each plant, you can create organisational models for your plants.

Cost Elements

The Style Production company profile provides 14 cost elements for you to use to build up and analyse item unit costs. Ten elements are standard within Style and four elements are user-defined.

Defining Production Capacity

This section looks at the parameters and resources that describe the production capacity available in your environment.

Within this context, capacity is synonymous with time. That is,

1. How many working days are there available in a week and in a year?
2. How many hours does it takes to make a product?

When you have established these two factors, you can determine whether you have enough, too little or too much (that is, spare) capacity.

Production Calendars

A production calendar defines how many working days there are in a year. You can define one calendar as the company default to be applied on all occasions or you can define more specific calendars.

If there are variations in the number of days different machines and/or work centres operate within your environment, use multiple calendars. For example:

Calendar	Scenario	Working Days	Non Working Days
Calendar 01	Default calendar	Monday to Friday	Weekends, bank holidays and national holidays.
Calendar 02	For machines that cannot stop, such as knitting machines	Every day	-
Calendar 03	A particular work centre produces swimwear in anticipation of the summer season.	Monday to Friday Weekends during January, February and March	Bank holidays and national holidays. Weekends from April through to December.
Calendar 04	Every 6th week a particular machine is shut down for maintenance.	Weeks 1 to 5, 7 to 11, ...	Weeks 6, 12, ...

Machine Capacity

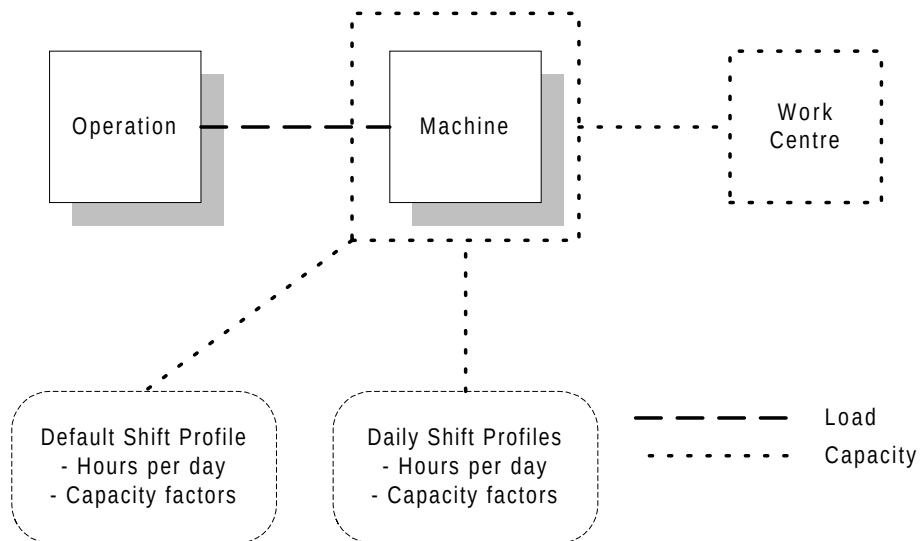


Figure -: Machine Capacity & Load

First of all, decide whether you wish to set up machines as single or multiple resources:

- Single Resource. Define each machine individually.
- Multiple Resource. Group machines of a similar type into a work centre. For example, you may wish to group all your knitting machines as a multiple resource.

You must book each operation on a production route to one machine or one work centre.

To load work across a number of machines without grouping those machines into a work centre:

1. Define a separate route for each machine.
2. Load production orders manually to those routes.

You can either specify the hours for each machine directly or allocate a shift profile.

Shift Profile

A shift profile is simply a list of start and finish times, identified by numbers, for each work shift.

When creating a shift profile, you can enter a new shift pattern after the current pattern to come into force on a specified future date.



Note: There is no default profile for a working day, this must be explicitly defined against the machine and shift profile codes.



Warning: The end date of the calendar is determined by the start date entered and the number of days defined in the Calendar section of the Company Profile. Therefore, set the Company Profile to allow creation of calendars for 52 and 53 weeks.

Labour Capacity

To be able to review labour load, you should:

- Define departments.
- Define labour skills.
- Define labour profiles, combining labour skills.
- Assign labour profiles to operations.

Style can then:

- For each operation, determine a set up time and an operation time.
- Determine the length of time each skill level is required at set up and run time and the required number of operators.
- Determine capacity requirements at department and skill level.

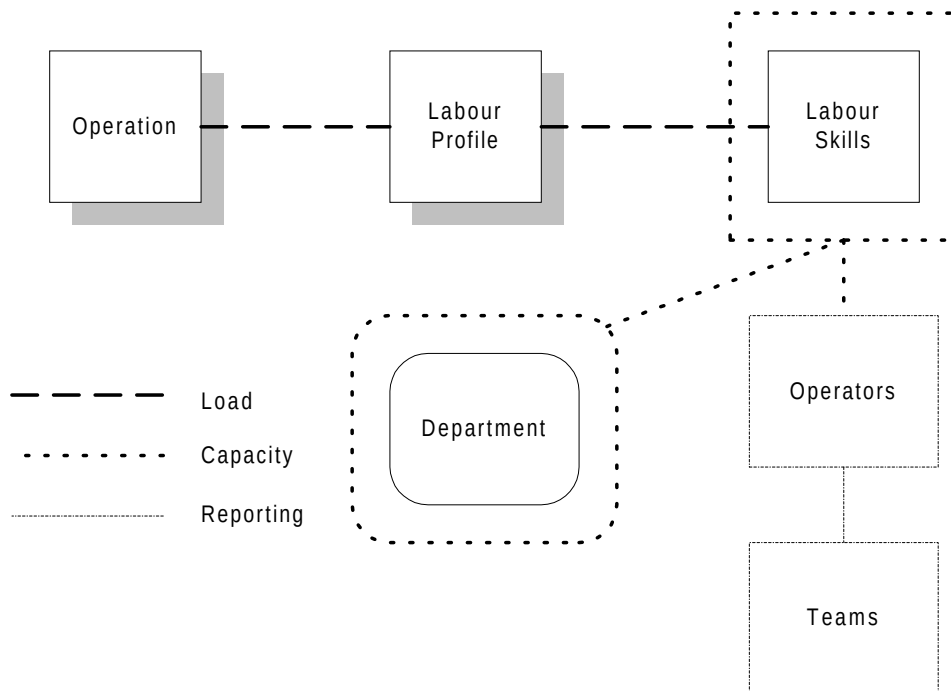


Figure -: Labour Capacity & Load

Labour hours per day are held by department, along with absenteeism, utilisation and average performance percentages.

Load is calculated at labour skill level by using the labour run times and the labour profile attached to an operation.

Capacity is calculated by:

- Adding up the operators attached to a labour skill and department on a given day.

- Using the standard hours per day, absenteeism, utilisation and standard performance values associated with the department in the following calculation:

$[\text{Std Hrs/Day} - (\text{Std Hrs/Day} * \text{Absenteeism})] * \text{Utilisation} * \text{Average Performance}$

Operations can be reported at operator or team level.

Scheduling Work

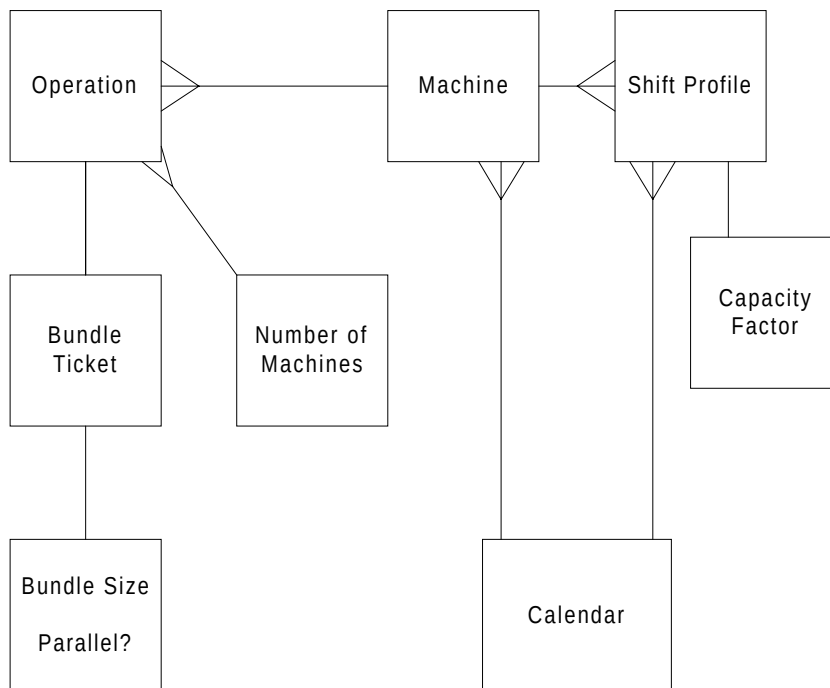


Figure -: Scheduling Work

Work is scheduled using the following procedure:

Firstly, calculate the available shift hours per day:

- Shift profile hours (effectivity date controlled).

OR

- Total of shift lengths.

OR

- Default to 8 hours.

Secondly, calculate the number of machines to be used in parallel; this is the lowest of:

- Number of machines on operations.
- Number of bundles on order:

$$\left(\frac{\text{Order Quantity}}{\text{Default Bundle Quantity}} \right)$$

- Capacity factor if shift profiles exist.

$$\left(\frac{\text{Shift Capacity}}{\text{Shift Length}} \right)$$

Thirdly, if no shift profiles exist:

- Calculate available hours on machine for the order

(Available Shift Hours * Number of Machines)

Machines

You can define a machine in either of the following ways:

- As an individual resource.
- To represent a logical view of total resource capacity.

Capacity analysis looks at machines to establish whether they have sufficient capacity to fulfil a proposed production schedule.

Machines may be assigned to:

- Operations. You need to associate either a machine or a work centre with each operation. Then, Style can develop work schedules and identify the resources that are available to carry out the work.
- Work Centres. Optionally, use work centres to group machines for different views of capacity requirements and constraints.



Tip: You can create a summary capacity route for a style, with each operation linked to a machine that represents the total capacity of a group of similar resources.

Standard And Maximum Capacity And Efficiency

You can define a machine's standard capacity and maximum capacity:

- Standard capacity indicates the default number of hours available on a working day. That number may vary between different days in the week.
- Maximum capacity indicates the number of hours that can be made available on a day. For example, an extra shift might be introduced or a shared or redundant resource might become fully available. The maximum capacity is only effective on working days defined on the production calendar. To introduce additional days, such as weekend working, you would need to change the current calendar or apply a new calendar.



Note: Capacity planning uses standard capacity when scheduling work at a machine. To reschedule using maximum capacity, you must change the standard capacity values and re-run the scheduling process.

Taking Account Of Performance Differences

Individual machines vary in performance. When comparing scheduled load to available capacity, you can apply a standard efficiency factor. You can use this factor where there are noticeable performance differences between machines, especially in groups or cells.

For example, if you have defined a standard operation time of 60 minutes, then machine 001 may actually take 72 minutes to complete the operation and machine 002 may take only 55 minutes. An efficiency factor can take account of machine 002 completing ahead of machine 001 when processing the same work.

Calculating Hours Per Day Per Machine

A production calendar associated determines the number of days a machine is operational throughout the year. By default, this is the calendar on the Company Profile.

There are four methods of calculating the number of hours available on working days. These methods are listed below in ascending order of complexity:

- 8 hours default.
- Shift details.
- Default shift profile.
- Shift/capacity profile.

The following tables illustrate, in ascending order of complexity, the four methods of defining working hours for machines over one working week.

- The examples use a default production calendar (01).
- A thick line border encloses the values set for each machine.
- The Std Capacity and Max Capacity columns indicate the resulting machine capacity on a day to day basis.
- The totals boxes at the foot of each table show the standard and maximum capacity of the machine calculated over one working week.

8 Hours Default

If no time is defined for the machine, Style assumes a default of 8 hours for both standard and maximum capacity for every working day within the year.



Note: You cannot use this method for multiple resources.

Production Calendar - 01				Machine A			
Date	Work Day?			Std Hours	Std Capacity	Max Hours	Max Capacity
Jan 1 Mon	Y			8	8.00	8	8.00
Jan 2 Tue	Y			8	8.00	8	8.00
Jan 3 Wed	Y			8	8.00	8	8.00
Jan 4 Thu	Y			8	8.00	8	8.00
Jan 5 Fri	Y			8	8.00	8	8.00
Jan 6 Sat	N						
Jan 7 Sun	N						
					40.00		40.00

Shift Details

You can define a standard and maximum capacity for up to three shifts per working day. When totalled, these give the number of hours available per working day.

In addition to shift lengths, you can also define a standard and maximum capacity that applies to every working day within the year. This is used to indicate that a machine represents more than one resource.

For example, if Machine A represents 2 sewing machines that operate 8 hours per day, then the standard capacity is 16 hours.



Note: With this method, you have to change capacity availability individually for each machine..

Production Calendar - 01				Machine A			
Date	Work Day?			Std Hours	Std Capacity	Max Hours	Max Capacity
				8	16.00	16	32.00
Jan 1 Mon	Y			8	16.00	16	32.00
Jan 2 Tue	Y			8	16.00	16	32.00
Jan 3 Wed	Y			8	16.00	16	32.00
Jan 4 Thu	Y			8	16.00	16	32.00
Jan 5 Fri	Y			8	16.00	16	32.00
Jan 6 Sat	N						
Jan 7 Sun	N						
				80.00			160.00

Default Shift Profile

You can apply a default shift profile to every working day within the year. In addition, you can allow for multiple resources or additional time by specifying a standard and maximum capacity factor. These factors uplift the number of hours defined by the profile per day by the required amount.

The number of hours per day are shown as effective hours. If the shift profile is made up of variable shift lengths, then effective hours may vary.



Tip: The advantage offered by this method is that, if there are changes in shift length, you simply update the shift profile rather than updating each machine.

Production Calendar - 01				Machine A			
Date	Work Day?	Default Shift Profile	Effective Hours	Default Std Factor	Std Capacity	Default Max Factor	Max Capacity

		1		1.23		2.45	
Jan 1 Mon	Y		8		9.84		19.60
Jan 2 Tue	Y		8		9.84		19.60
Jan 3 Wed	Y		8		9.84		19.60
Jan 4 Thu	Y		8		9.84		19.60
Jan 5 Fri	Y		8		9.84		19.60
Jan 6 Sat	N						
Jan 7 Sun	N						
					49.20		98.00

Shift/Capacity Profile

You can allocate one standard shift profile as the default, together with specific shift profiles for those days that deviate from standard.

The example in the table below allocates three shift profiles to Machine A:

Shift Profile	Description of Shift
1	Standard 8 hour shift
2	Friday shifts with an earlier finishing time
3	Shift pattern for weekend working

In addition, you can apply a standard and maximum capacity factor to each day of the week. For example, you may wish to cater for fewer machines running at weekends. Similarly, you can set up a default code for days when nothing specific is required. Also, you can use effectivity dates to control the number of hours per day.

This method provides the most exacting control on capacity calculations. Again, by changing the profiles you change the daily effectivity hours. Capacity factors, however, should be changed at machine level.

Production Calendar - 01				Machine A			
Date	Work Day?	Shift Profile	Effective Hours	Std Factor	Std Capacity	Max Factor	Max Capacity
Jan 1 Mon	Y	1	8		8.00	2.34	18.72
Jan 2 Tue	Y	1	8		8.00	2.56	20.48
Jan 3 Wed	Y	1	8	1.23	9.84	2.56	20.48
Jan 4 Thu	Y	1	8		8.00	2.56	20.48
Jan 5 Fri	Y	2	6	1.1	6.60	1.98	11.88
Jan 6 Sat	N	3	4		4.00		4.00
Jan 7 Sun	N	3	4		4.00		4.00
					48.44		100.04

Bundle Tickets

A bundle ticket attaches to each bundle (bag, garment etc.) on a production order.

Style can print bundle tickets automatically, for example at order release. Otherwise, you can request Style to print bundle tickets via a menu option.

An example of a bundle ticket is given below:

Bundle Ticket	
Style NJACKET Colour SKY	Blue Size 36 *REPRINT*
New Jersey Jacket	Order W000212
Ticket Jacket fronts	Quantity 12
Bundle No. 0001516	Due Date 12/05/96
Lot Number JBAD44	Next Machine MK12 Next Dep MAKU
0020 First front operation	0030 Second front operation
Optional bar code	Optional bar code
000015160020 6.2750	00015160030 3.1575

The bundle ticket includes basic data such as:

- Production order number.
- Style.
- Bundle number.
- Colour and size of garments in the bundle.

An operation stub is attached for each operation that is to be performed on the bundle. This stub includes the following information:

- Operation sequence.
- Description.
- Barcode.
- Standard minutes.

The bundle may also include a bar code for speed and accuracy of feedback.

Bundle Ticket Types

Within Style Production, you can create bundle tickets for different purposes. The following circumstances require different types of bundle tickets:

- Bundles with varying quantities.
- Dividing the route into sections.
- Parallel processing.

Examples are given below.

Bundles with varying quantities

When producing socks, different bundle ticket types may be issued for:

- Knitting (120 pieces per bundle).
- Dyeing (750 pieces per bundle).
- Finishing (240 pieces per bundle).

Dividing the route into sections

In a cut and sew environment, a cutting instruction may be issued to a style. Depending on the mix of sizes and cloth widths, the actual quantities cut may vary from the instruction.

Therefore, all following operations need to be based upon the actual cut quantities and not the original order quantities.

Parallel processing

When producing a shirt, separate components may be produced simultaneously on different machines in various factories.

Parallel Operations			
Operation		Bundle Ticket	Duration
0010	CUT	n/a	2 days
0020	Front 1	FRT	1 day
0030	Front 2	FRT	2 days
0040	Front 3	FRT	1 day
0050	Back 1	BCK	1 day
0060	Back 2	BCK	2 days
0070	Back 3	BCK	1 day
0080	Sleeve 1	SLV	1 day

0090	Sleeve 2	SLV	1 day
0100	Cuff	CUF	1 day
0110	Make up	MUP	7 days
0120	Pack		1 day

Fronts, backs, sleeves and cuffs may be produced separately and gathered together at an assembly point. In this case:

- A separate bundle ticket can be attached to each bundle of fronts (FRS), backs (BCK), sleeves (SLV) and cuffs (CUF).
- The operations associated with each group of bundles (that is, fronts, backs etc.) can be scheduled to run in parallel.
- These operations can then be brought together at a marry up (MUP) point.



Note: To use parallel processing, you must use bundle tickets that have had the Parallel Ticket Type set to 1 on the Maintain Bundle Ticket screen.

Parallel Processing					
Day	Operations				
1			0010 Cut		
2			0010 Cut		
3	0020 Front 1	0050 Back 1			
4	0030 Front 2	0060 Back 2			
5	0030 Front 2	0060 Back 3		0080 Sleeve	
6	0040 Front 3	0070 Back 3		0090 Sleeve	0100 Cuff
7-13			0110 Make up		
14			0120 Pack		

By using different bundle ticket types for different operations, further analysis groups operations by ticket type to determine the actual number of processes run in parallel. The critical path through these operations is the longest lead time associated with a grouped parallel ticket type.

Style Production Routes

Style Production provides an extremely versatile method of defining style production routes.

The design of the application has been influenced by the need to link materials and operations in order to obtain an integrated model of product and process. This integrated structure allows you to plan, produce and cost products with great precision and control.

For each style production route, you define:

- Route. A series of consecutive operations in which materials are transformed into a finished product, to be received on completion into a stockroom or warehouse.
- Bill of materials. A list of materials. Link each material to a specific operation, in order that MRP can plan when you need the material.

Each style must have at least one production route.

However, you can set up multiple routes for one style. Examples of when you may require more than one route for a style are given below:

- To contract out certain operations.
- To maintain separate costing and planning routes.
- To define summary route/BOMs for rough cut capacity planning.
- To take account of capacity constraints or alternative processes.
- To accommodate different batch sizes where economies of scale may influence the routing procedure.
- To model different costing strategies.



Tip: Routes are the fundamental building blocks of Style Production. Invest as much time and effort as possible to ensure that they are accurate, complete and kept up-to-date.

Building A Style Production Route

You can build a route/BOM by:

- Inputting operations manually.
- Copying in operations from groups of standard operation that you set up beforehand.
- Copying in operations from existing routes for the same style or a different style.

Once copied, you can maintain operation details separately for the copied-to product.

Within Style Production, you define production routes at style level. By doing so, you save a lot of time not having to define a production route for every style variant.

You specify which, if not all, of the style's colours and sizes you wish to produce on a particular route. If an individual colour/size or group of colours/sizes uses a variant of the standard route, you can either build a separate route or you can specify variant operations on the same route.

You can define lot size routes that allow you to define product specifications in terms that different functions within your organisation will find appropriate. For example, different roles within the company may build different routes as shown below:

- Accountants in terms of the style's economic production batch size or economic order quantity (EOQ).
- Production planners in terms of the optimum production batch size.
- Material planners in terms of the natural replenishment quantity for the style.

You can control these different views of the style by specifying designated routes for material planning and style costing to ensure consistency of calculation.

Because product specifications and process methods are logically tied together, information is shared between functional areas. For example, industrial engineers and product designers can be encouraged to work together when developing a new style route, and a production design is more likely to result.



Tip: To avoid a situation where numerous unofficial routes/BOMs are held independently of one another, ensure that every functional department has a stake in the route data. This should encourage communication and a greater cohesion between departments, thereby promoting the benefits of information sharing.

Standard Operation Groups

To save time rekeying similar operation definitions against multiple routes, you can define standard operation groups.

Standard operation groups may include both primary and variant standard operations. You can copy standard operations as primary or variant operations to a live production route or model planning route.

Through a unique Specification Id, you can retain a link between route operations and the standard operations from which they originate. Then, when you make changes to a standard operation, you can perform a mass update on the route operations.

There is a parameter you can set up to control which fields get updated during a mass update.

Extended Text Descriptions

You can use an extended text description to enter information for each operation. This information may be essential for complex or commercially sensitive processes, where additional technical or procedural instructions are necessary.

- In support of a quality improvement programme, you can arrange for this information to print out on production order documentation.

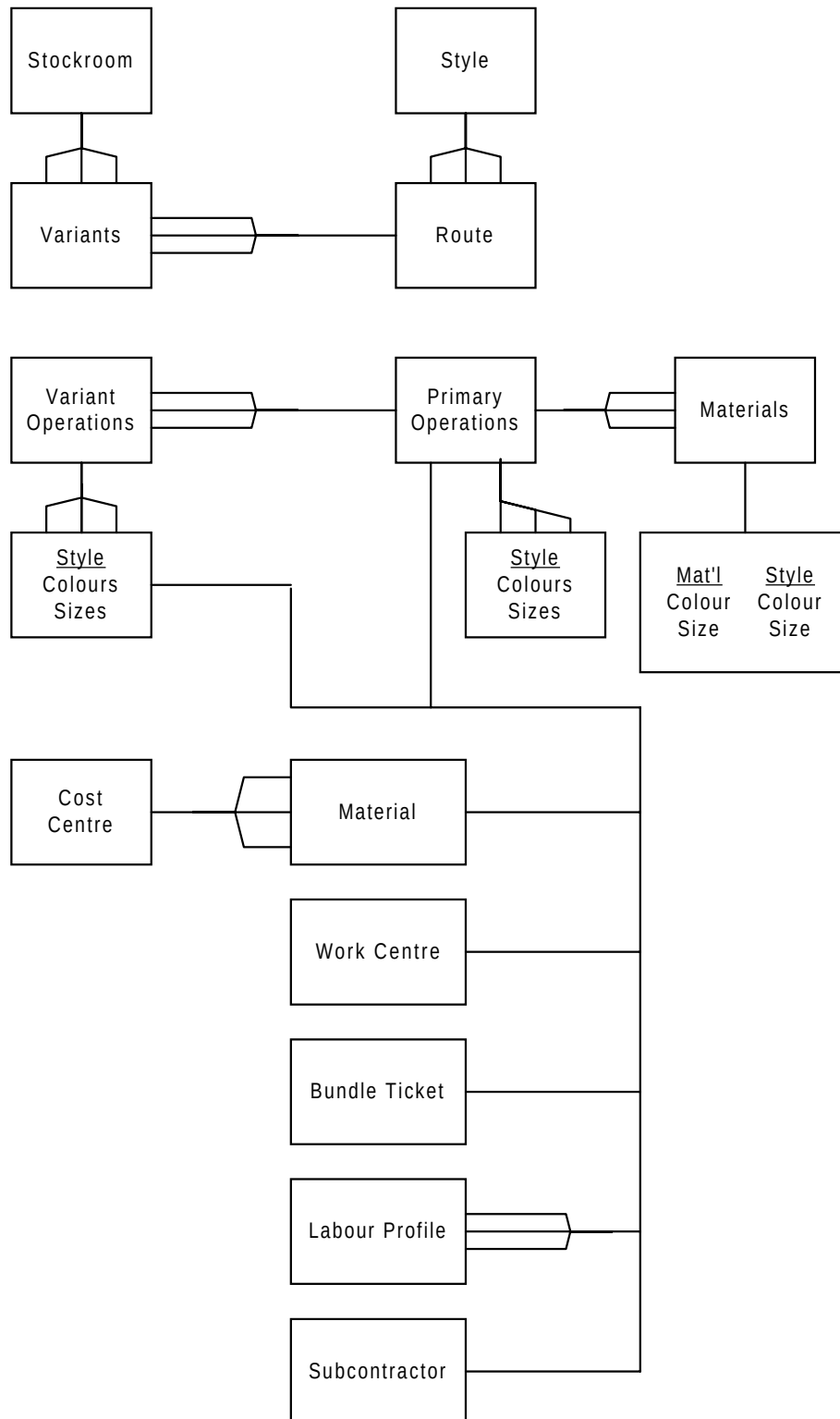


Figure -: Route/BOMs Database Relationships

The above figure shows how a route/BOM relates to materials and resources. A trident indicates a one to many relationship.

Materials

When you define a route, you provide details of the materials required to make styles and individual style variants.

Styles may require common materials for all style and colour variants. For example, you may make shirts in blue, green and red variants. Each variant may require the same fabric, 100% cotton, but in different colours. In this case, you can use a matrix to relate the material characteristics to the style characteristics. For example, you can specify green fabric for a green shirt.

Alternatively, you can specify different materials for individual colour and/or size variants.

In addition, the material quantity requirements may vary between variants. This is usually associated with larger sizes requiring more material than smaller sizes.

To this end, Style can calculate material requirements automatically for all variants, when you define the quantity requirement for a primary variant and then specify a ratio to determine the remaining variant requirements.

Alternatively, you may wish to specify each requirement individually; with the system defaulting to the standard material requirement entered when first defining the material code against the operation.

Updating Material Details On Routes

To identify and update materials on a route where-used basis, you can use the Mass Replace Materials option. This option saves you the trouble of identifying and updating every occurrence individually.

Operations

There are two types of operation that you can define on a route:

- **Primary.** A primary operation is an operation used by default to produce the style variants. Style uses primary operations to cost the style.
- **Variant.** A variant operation is an operation that is specific to a certain colour/size variant of a style. The operation details may vary significantly from those defined for the primary operation.

For each primary or variant operation, you need to give the following information:

- Where the operation is carried out.
- How long the operation takes to complete.
- What materials the operation uses.
- How to report completed work.
- The machine at which the operation takes place.

This information is necessary for determining the time-phased load created by the production plan and calculating anticipated production costs.

Lead Time And Cost

For each operation, you specify the standard machine and labour time that it takes to complete the operation. You can base machine and labour time on one of the following units:

- Time taken to produce a standard batch.
- Time taken to produce a quantity of 1.
- Time taken to produce a quantity of 100.
- Time taken to produce a quantity of 1000.

This information is important for two reasons:

- To determine the lead time of the operation for scheduling.
- To cost the operation.

Other factors involved in calculating lead time are:

- **Number of machines likely to be available to perform the work.** For example, if time is 10 hours but there are 2 machines, then the lead time is only 5 hours.
- **Move days factor.** This factor indicates any excessive time lost in transferring work from a previous operation. You can add in this factor at machine level by assigning a queue time to the machine.

The cost of the operation is also further qualified by team size or labour profile which determine the number of people required for the operation. Costing routines multiply the number of people by their charge rates to arrive at the labour cost involved in production.

Both costing and lead time values may further be inflated if there is any wastage percentage associated with the operation. This is increased time spent at the operation due to inefficiency. (In capacity planning this may be off-set by machine and labour efficiency factors).

Variant Operations

The information that you enter for a primary operation applies to all variants of the style. However, this is unrealistic in certain industries that may define a route to cater for a large number of colours/sizes, for example, children through to adults. The time taken to complete one or a batch of styles of children's' sizes is less than that for adults.

To record varying operation details for specific colours and/or sizes, you can set up one or more variant operations. Variant operations run parallel to a style route operation.

From a primary operation, you can define one or more variant operations for certain colours or sizes or colour/size combinations.

Use variant operations to specify:

- Alternative standard production times.
- A different machine.
- A different labour profile (that is, skill mix).
- Changes to required materials, covering quantity per size and matching the colour of the style to that of the material.

Calculating Operation Lead Time

Operation lead time is the time taken to complete an operation. Lead time is critical for the following functions:

- Scheduling and planning. Style uses the lead time for each operation to determine when to start a production order in order to achieve the due date. This process involves fitting the number of hours required into the number of hours available at the appropriate machine on each working day.
- Capacity loading. Style reviews the hours available per day with that required to progress an order.
- Costing. Style multiplies the machine lead time and the labour lead time by a rate to obtain a cost.

To calculate lead time for an operation, Style uses the following formula:

$$\text{Operation Lead Time} = \text{Queue Time} + \text{Operation Duration} + \text{Move Time}$$

where:

Queue time Is the time that completed WIP inventory from a previous operation must wait before work can start on the current operation.

Operation duration Is the time that WIP is worked with or on during the current operation.

Move time Is used generally to indicate long inter-operational delays, for example moving work between sites.

Standard Operation Lead Time						
Op	Queue Time	Operation Duration	Move Time			Total Lead Time
	Hrs	Hrs	Days	Std Shift Length	Hrs	Hrs
10	2.0000	4.0000	1.000	8.000	8.0000	14.0000
20	3.0000	5.0000	0.000	8.000	0.0000	8.0000
30	4.0000	6.0000	3.000	8.000	24.0000	34.0000
40	2.0000	3.0000	0.000	8.000	0.0000	5.0000
	11.0000	18.0000	4.0000		32.0000	61.0000

Operation Duration Time

Operation definitions are very flexible in order to mirror the wide variety of methods that may be employed to calculate duration time.

For the purposes of scheduling, operation duration may not be straightforward. The work load may be split between more than one machine.

In addition, it is possible to enable a series of operations to run concurrently (that is, parallel processing) that will reduce the actual lead time dramatically as opposed to completing each operation in turn. However, this does not affect the duration time of the operation itself; nor does it affect costing routines as machine and operator time requirements remain the same.

Standard operation duration depends firstly on manually keying in the following information per operation:

- Set up labour time.
- Run labour time.
- Run machine time.

The exact meanings of these times, entered in hours or minutes depending on the company default requirement, are not necessarily immediately obvious. For example, if you enter a run labour time of 1 hour at an operation, what does it mean? What does your labour produce in that time?

In order to clarify this, a series of time basis codes are available. These codes allow you to clearly indicate what exactly that 1 hour means. You must enter one time basis code per operation, but each operation is independent in its choice of codes.

Time Basis						
Lot	Time Basis Code			Set Up Labour	Run Labour	Run Machine
	Description	Batch Qty	No.			
				Hrs	Hrs	Hrs
1000	0	Time per parent lot		1.1000	12.0000	12.0000
1000	1	Time each		1.1000	0.0050	0.0065
1000	2	Time per 100		1.2300	2.3330	1.9800
1000	3	Time per 1000		1.3400	4.5600	3.5000
1000	4	Quantity per hour (n/a)				
1000	5	Fixed time operation		1.4500	7.8000	12.0000
1000	6	Time per operation batch	500.000 2	1.3300	1.6780	0.9600

Another point to consider when calculating operation duration is: do set up labour time, run labour time and run machine time run alternately or are they concurrent?

You may for example, have a run labour time of 5 hours and a run machine time of 7 hours; but the labour time is a 5 hour requirement within the 7 hours of machine time.

Your operation duration time would therefore be set up time plus 7 hours. However, you would still need the run labour time to cost the operation correctly.

Or, for example, you may have a set up time of 2 hours, a run labour time of 3 hours and a run machine time of 4 hours; where the labour component must be carried out before the machine component can start. In this circumstance, you would want an operation duration time of set up plus run labour plus run machine.

To determine exactly how to combine the three time components, Style provides a series of duration basis codes:

Duration Basis						
		Duration Basis Code				
		0	1	2	3	4
Run Machine Time	Run Labour Time	Set Up Only	Set Up + Run Machine	Set Up + Run Labour	Set Up + Run Labour + Run Machine	Set Up + greater of Run Labour or Run Machine
Hrs	Hrs	Hrs	Hrs	Hrs	Hrs	Hrs
2.00	2.00	1.00	3.00	3.00	5.00	3.00
10.00	7.00	2.00	12.00	9.00	19.00	12.00
3.50	1.25	1.50	5.00	2.75	6.25	5.00
20.00	15.00	3.00	23.00	18.00	38.00	23.00

Wastage

You can increase the duration time in order to compensate for expected wastage. This increases the WIP inventory requirement at each operation in order to achieve the operation target.

This wastage % is specified at operation level.

Wastage						
Run Qty	Std Lot Size	Std Duration	Wastage	Cumulative Wastage Factor	Inflate Run Qty	Inflate Standard Duration
100	100	Hrs	%			Hrs
Operation 30		5.0000	20	1.25	125.0	6.2500
Operation 20		8.0000	5	1.32	131.6	10.5263
Operation 10		3.0000	10	1.46	146.2	4.3860
					w2	
		16.0000				21.1623

Subcontract Operations

Certain processes involved in the production of styles may be carried out off-site at subcontractors. You can include these processes as operations on route/BOMs.

Style Production makes it easy to track work sent to and received from subcontractors. You can track and monitor the:

- Movement of materials to a subcontractor.
- Transfer of WIP inventory to a subcontractor.
- Progress of WIP inventory at a subcontractor.
- Receipt of WIP inventory from the subcontractor.

The steps involving subcontracted operations are as follows:

1. Set up the system.
2. Set up Route/BOM.
3. Release the production order.
4. Create purchase orders for all subcontracted operations.
5. Allocate lots.
6. Issue materials.
7. Report preceding operation.
8. Ship WIP to the subcontractor.
9. Progress at subcontractor (if applicable).
10. Receive from subcontractor.
11. Create GRN for purchase order.

System Set Up

Subcontracting depends upon Style Purchase Management and Accounts Payable being live. Each subcontractor must be a valid supplier. Purchase orders may be raised automatically for all subcontracted operations.

You must define a subcontractor stockroom on the Organisational Model for holding all material balances for materials required by the subcontractor to complete the operation(s).

Set up the Shipper Tracking flag under the Organisational Models option to do one or both of the following:

- Track the movement of WIP inventory and/or materials through a shipper number. Style assigns a number to each shipment. You can, however, use the same number to track amounts of WIP from more than one production order or multi shipment within the same order.
- Print shipper notes (that is, despatch notes) to accompany each shipment.

Route/BOM Set Up

You can define primary operations and variant operations as subcontracted operations.

A subcontracted operation cannot be the first operation on a route/BOM. If you need to get round this, create a dummy count point operation at which to register the work-in-progress to ship.

You can record standard labour, machine and set up time to take into account the capacity and loading of subcontracted work when planning.

During material issuing, Style transfers materials to the subcontractor stockroom defined on the Organisational Model. When the operation that the materials are attached to is booked by either the Progress at Subcontractor or Receive from Subcontractor routines, Style backflushes the materials out of the stockroom.

Material balances are held by subcontractor/material within the stockroom.

Order Release

The Release Orders option identifies subcontracted operations on the production route. This option generates purchase orders automatically for all operations where you set the Key Operation field to 2 or 3.

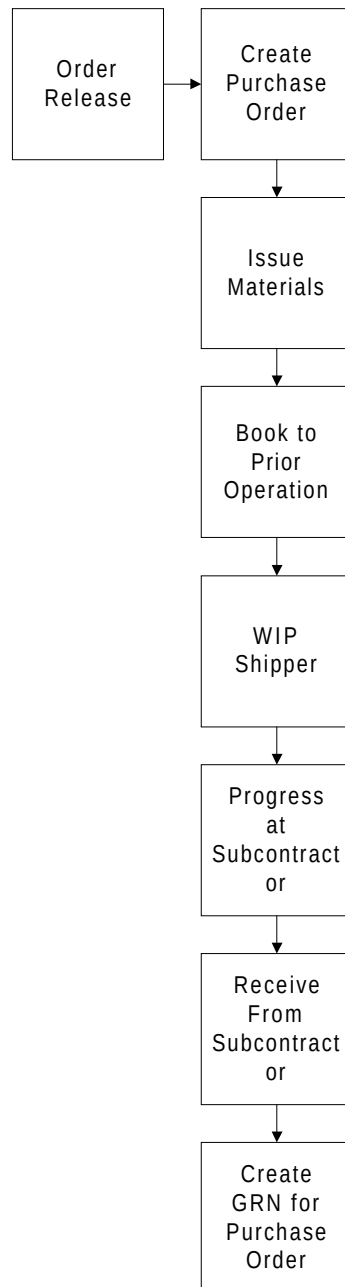


Figure -: Subcontract Work Cycle

Create Purchase Orders

Purchase orders are created automatically for the supplier defined on the production route. A purchase requisition is not necessary.

Allocate Lots

If any of the materials to be shipped to the subcontractor are batch or lot controlled, then these materials must be allocated to the production order. Style can then transfer the correct lots to the Subcontractors Stockroom. Later on, these batches/lots can be backflushed.

Issue Materials

Issue any materials required by the subcontractor. Style transfers the materials from the issuing stockroom to the subcontractor stockroom defined on the Organisational Model.

Report Preceding Operation

This is necessary in order to ascertain the quantity of Available WIP to move to the subcontractor.

WIP Shipper

Work-in-progress is now shipped to, and optionally tracked at, the supplier. This option reduces the Available WIP quantity balance at the prior to subcontracted operation, and increases the subcontractor WIP quantity balance at the same operation with the same quantity. Enquiries on subcontractor WIP inventory balances must therefore be made on the prior operation and not the subcontracted operation.

Progress At Subcontractor

If consecutive operations are subcontracted, then it is possible to record bookings at each operation as if the work was being carried out on-site.

Also, subcontractor material issue balances are downdated with material quantities. At the point of receipt, you may override these quantities.

Receive From Subcontractor

On receipt, Style:

- Increases the Available WIP inventory balance at the subcontracted operation by the quantity returned.
- Decreases the subcontractor WIP balance at the prior operation by the same quantity.
- Downdates subcontractor material issue balances with material quantities. At the point of receipt, you may override these material quantities.

Quantities returned from a subcontractor can be received into held status and returned for rework if necessary; subcontractor and prior operation WIP inventory balances are updated appropriately.

Create GRN For Purchase Order

It is customary for Style Purchase Management to carry out this function. Within Style Production, you can report receipts without a GRN (goods received notes) if a different purchase order to that raised by the release procedure is entered during receipts processing.

How Costing Is Performed

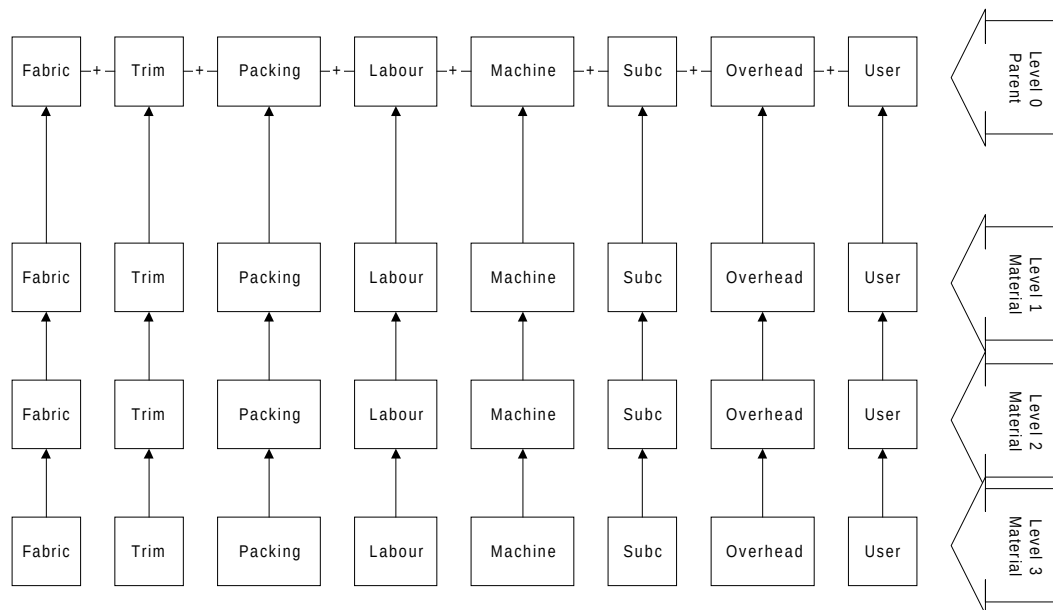


Figure -: Cost Rollup

Cost elements are rolled up into higher level costs to arrive at a full product cost. These cost elements indicate production, materials and user-defined costs at each level in a production route/BOM:

- Production costs comprise labour (set up and run), machine, subcontract and overheads.
- Material costs may be subdivided into fabric, trim and accessory costs.
- User-defined costs may be entered manually; or may comprise exclusively or include landed costs.